

YILI YANG, PhD

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AI Scientist | Geospatial & Remote Sensing

SKILLS

DOMAINS

Geospatial Information Science
Remote Sensing
Machine Learning & Deep Learning
Image Segmentation
Object Detection | Multimodal Learning
Time-series modelling
Earth Foundation Models

TECH STACK

PyTorch | Scikit-learn
Google Earth Engine | QGIS
Google Cloud Platform | ArcGIS Pro

ENGINEERING

Docker | GitHub Actions
Git | GitHub

EDUCATION

UNIVERSITY OF EDINBURGH

PHD PETROPHYSICS, 2021

MSc GEOLOGY, 2016

BSc GEOSCIENCE, 2015

STATS

PUBLICATIONS

Journal Papers: 13

Conferences: 8

Data Sets: 1

ACADEMIC SERVICES

Invited Talks & Panels: 17

Peer-reviews: 8

Mentees: 9

LINKS

Personal Website

LinkedIn || ORCID

Google Scholar || GitHub

EXPERIENCE

DATA SCIENTIST | WOODWELL CLIMATE RESEARCH CENTER

Jan 2022 - Present | Falmouth, MA, USA

- **Project Lead:** Pan-Arctic thaw-slump mapping — deep learning on multi-source satellite imagery to detect and quantify features at continental scale
- **Sensor fusion:** Integrated multiple satellite imagery types with field observations into ML workflows, improving feature-detection accuracy
- **Production scale:** Optimised deep-learning model inference over Arctic-scale geospatial data on Google Cloud Platform
- **Geospatial analytics:** Produced pan-Arctic statistics from large-scale geospatial datasets
- **ML modelling:** Computer-vision wildfire mapping and machine-learning wildfire prediction from environmental variables (classification & regression)
- **Time-series:** Carbon-flux time-series imputation and multi-domain climate time-series prediction
- **Foundation models:** Applied vision foundation models (SAM, location embeddings) to satellite image segmentation

DATA SCIENCE FELLOW | FACULTY.AI

Sep 2021 - Dec 2021 | London, UK

- Professional data-science fellowship in machine learning and AI with an industry client placement

DOCTORAL RESEARCHER | ICCR

International Centre for Carbonate Reservoirs

Sep 2016 - Apr 2021 | Edinburgh, UK

- Computational methods for multiphase fluid flow; deep-learning image segmentation for digital rocks
- Synchrotron X-ray μ -CT imaging and analysis

AWARDS & GRANTS

FUNDING FOR CLIMATE SOLUTIONS

Funding for Climate Solutions, Woodwell Climate Research Center
\$99,749 | 2025-2026 | A Generic Climate AI Framework for Multi-domain Time Series Prediction

MULTIDISCIPLINARY COLLABORATIONS

GOOGLE × WOODWELL

Understanding Permafrosts, AlphaEarth and Satellite Embedding

ESRI × WOODWELL

Understanding Permafrosts and Indigenous Communities

ACADEMIC COLLABORATIONS

University of Connecticut | Arizona State University | University of Illinois